

Practical-4

JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Theory

1. How are DOM elements selected and modified?

Answer: DOM elements are selected using methods like `getElementById()`. They are modified using properties like `textContent`, `style`, and `classList`.

2. Are event listeners attached correctly?

Answer: Yes, event listeners are attached using `addEventListener()` for actions like button clicks, form submission, and menu toggling.

3. Is `localStorage` used for remembering theme choice?

Answer: Yes, `localStorage` is used to save the user's selected theme so that the theme remains the same after refreshing or reopening the website.

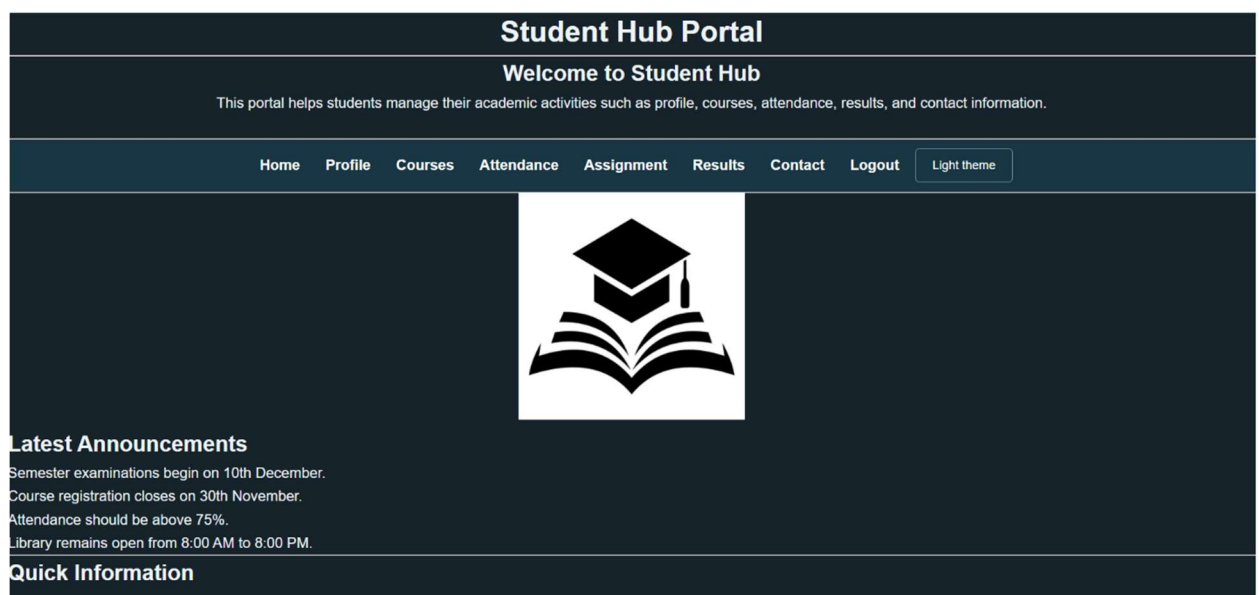
4. Does interactivity improve usability without breaking accessibility?

Answer: Yes, features like form validation, theme switching, notifications, and menus improve usability while maintaining accessibility through proper buttons, labels, keyboard support, and readable colors.

A. Result:-

CSS Code:-

Evidence of Work (Screenshots)



Feature	Description
Profile	View and update your personal details.
Courses	View your enrolled courses.
Attendance	Check your attendance percentage.
Results	View semester examination results.
Contact	Send your questions or feedback.

Student Hub Portal
Email: studenthub@college.edu
Phone: +91 9876543210

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[Home](#) [Profile](#) [Courses](#) [Attendance](#) [Assignment](#) [Results](#) [Contact](#) [Logout](#) [Light theme](#)

Student Profile

Personal Information

Full Name: Krish
Username: Krish019
Gender: Male
Date of Birth: 05 September 2007
Blood Group: A+

Contact Information

Mobile Number: +91 9874653210
Email ID: krishdesai301@gmail.com
Address: Ahmdabad, Gujarat

Academic Information

Student ID: 25DCE019
Course: Computer Engineering
Year: Second Year
Semester: Third Semester
Division: A
Roll Number: 19
College Name: Charotar University Of Science and Technology

[Back to Dashboard](#)

Student Course Portal

My Courses

Course Code	Course Name	Faculty	Credits	Semester
CS101	Programming in C	Dr. Sharma	4	Semester 1
CS102	Computer Fundamentals	Prof. Mehta	3	Semester 1
CS201	Data Structures	Dr. Patel	4	Semester 2
CS202	Database Management System	Prof. Singh	4	Semester 2
CS301	Web Development	Dr. Kumar	3	Semester 3
CS302	Java Programming	Prof. Verma	4	Semester 3

Course Information

Attend all lectures regularly.
Complete assignments before the deadline.
Maintain at least 75% attendance.
Contact your faculty for any course-related questions.

[Back to Dashboard](#)

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Student Attendance Portal

To exit full screen, move mouse to top of screen or press and hold
Esc

Student Name

Enrollment ID

Branch

Select Branch

Semester

Select Semester

Attendance Record

Subject	Total Classes	Classes Attended	Attendance %
Mathematics	40	36	90%
Programming	45	40	89%
Database	38	35	92%
Web Technology	42	39	93%

Check Attendance

Clear

[Back to Dashboard](#)

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[Home](#)
[Profile](#)
[Courses](#)
[Attendance](#)
[Assignment](#)
[Results](#)
[Contact](#)
[Logout](#)

Light theme

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My Assignments

Assignment ID	Subject	Assignment Title	Due Date	Status
A101	Programming in C	Write a C Program for Student Record	10-Aug-2025	Submitted
A102	Database Management	Design Student Database	15-Aug-2025	Pending
A103	Web Technology	Create a Student Portal Homepage	20-Aug-2025	Submitted
A104	Java Programming	Develop Calculator Application	25-Aug-2025	Pending

Assignment Submission

Assignment ID

Subject

Upload File

Choose File
No file chosen

Upload File

Choose File

No file chosen

Remarks

Enter remarks

Submit Assignment

Clear

Instructions

Submit assignments before the due date.
Upload files in PDF, DOCX, or ZIP format.
Maximum file size: 10 MB.
Late submissions may receive reduced marks.

Back to Dashboard

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Home

Profile

Courses

Attendance

Assignment

Results

Contact

Logout

Light theme

Student Result Portal

Semester Result

Student Name

Krish Desai

Enrollment ID

25DCE019

Branch

B.Tech

Semester

Semester 3

Subject	Marks Obtained	Total Marks	Grade
Programming in Java	88	100	A
Database Management	91	100	A+
Web Technology	85	100	A
Data Structures	79	100	B+
Operating System	90	100	A+

Total Marks

433 / 500

Percentage

86.6%

Result

PASS

Grade

A

Back to Dashboard

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[Home](#) [Profile](#) [Courses](#) [Attendance](#) [Assignment](#) [Results](#) [Contact](#) [Logout](#) [Light theme](#)

Student Portal

Contact Us

Full Name	Enter your name
Email	Enter your email
Phone Number	Enter your phone number
Subject	Enter subject
Message	Write your message

College Contact Details

College Contact Details

Address: Depstar College, near vadtal, Anand
Email: krishdesai301@gmail.com
Phone: +91 9876543210

[Home](#) [Profile](#) [Courses](#) [Attendance](#) [Assignment](#) [Results](#) [Contact](#) [Logout](#) [Light theme](#)

Student Hub Login

Student ID:

Password:

[Register](#) | [Forgot Password?](#)

B. GitHub Repository Link:

<https://github.com/krish-301/Student-Hub-Portal>

C. Challenges Faced:

- • Understanding how to select and modify HTML elements using the DOM.
- • Handling different events like button clicks, form submission, and menu toggling.
- • Implementing the light/dark theme switcher using JavaScript and `localStorage`.
- • Making the hamburger menu, FAQ, popup, slider, and notification work correctly.
- • Maintaining proper styling and accessibility while adding interactivity.
- • Fixing errors and testing the JavaScript code using the browser console.

D. Learning Outcomes:

Learned how to select and modify HTML elements using JavaScript DOM methods.

Learned how to use event listeners to make web pages interactive.

Learned how to create dynamic features such as FAQ, popup, slider, notification, and hamburger menu.

Learned how to use `localStorage` to remember the user's theme preference.

Learned how JavaScript, HTML, and CSS work together to create an interactive website.

Improved debugging and testing skills using browser developer tools

Faculty Remark: